

5. Taxes as a Factor

Taxes clearly affect individuals' behavior, but do taxes affect asset prices? Originally there were two schools of thought. Miller and Scholes (1978—yes, the same Nobel Prize winning Scholes who asserted his own lack of tax expertise) said that although a given individual is affected by taxes, the average investor is not and taxes are not a factor. Intuitively, investors sort themselves into two groups: those with low tax rates, who tend to hold assets with high tax burdens such as high-dividend stocks, and highly taxed investors who hold assets with low tax burdens, such as stocks that pay no dividends.²⁵ In this way, taxes should become irrelevant for equilibrium pricing.

The other school of thought is that the representative investor is subject to taxes, and equity evaluations are affected (inversely) by tax burdens. Increases in tax burdens should be compensated by higher expected returns. The first model along these lines was by Brennan (1970). The empirical evidence has strongly settled the debate in favor of this camp. Taxes are a factor.

There are two ways taxes induce pricing effects.²⁶ There is a *dividend tax penalty*, where high dividend yield stocks are disadvantaged because of the higher rate of tax levied on dividends than on capital gains. In addition, stocks that have done well provide shareholders with accrued capital gains. Investors with too many shares of a stock that has appreciated in the past sell fewer shares than would otherwise be the case in order to avoid paying capital gains tax. There is no effect for investors buying because buying does not trigger capital gains. Thus prices of stocks with large accrued capital gains must be higher, and thus expected returns lower, to induce more selling by investors who are locked in. This is an example of *tax capitalization*, and it can lead to long-term reversals.

Sialm (2009) shows that tax effects exist in both the time series and cross-section of equity returns. Sialm computes an expected tax yield, which is the sum of taxes on dividends, short-term capital gains yields, and long-term capital gains yields. The capital gains yields component is small, as expected from the Constantinides (1983) tax-timing option. Sialm shows that when tax yields are high, aggregate market valuations are low. This is a time-series effect. In the cross section, Sialm shows that stocks with high tax yields (high dividend paying stocks) have higher returns. These returns are in excess of the Fama–French size, value, and momentum factors. This control is very important because stocks with high tax burdens are likely to have dividend payments, and these are more likely to be value stocks—which we know already have high returns (see chapter 7). Sialm

²⁵ Miller and Modigliani (1961) also predicted that investors in high tax brackets would be attracted to stocks with low dividend yields and vice versa. The paper does not focus on equilibrium with taxes; its major result is that the value of the firm is independent of the firm's dividend policy and was cited in the Nobel Prizes awarded to Modigliani in 1985 and Miller in 1990.

²⁶ This discussion follows Klein (2000).